Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/23/2006
PI Name:	Duval, Walter M.B. Ph.D.		
Project Title:	Stero Imaging Velocimetry of Mixing Driven by Buoyancy Induced Flow Fields		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44135	**Congressional District:**	10
Comments:	Suffix: referred to as Dr., assume it's a PhD -- rje 2/2/2017		
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	05/01/2000	**End Date:**	05/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Duval, Walter		
COI Name (Institution):	Batur, Celal (University of Akron)		
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
	Mixing driven by buoyancy induced flow fields will be addressed both experimentally and computationally. The experimental effort will address the kinematics of mixing: stretching, transport, and chaos. Quantification of the mechanisms of mixing will consists of measuring the flow field using Particle Imaging Velocimetry (PIV) and capturing the dynamics of the interface using a Charge-Coupled Devices (CCD) camera in order to quantify its Lagrangian history. These experiments will be carried out within the framework of Earth's gravity and eventually in a microgravity environment such as the International Space Station. The computational effort will corroborate the experimental findings with predictions of the dynamics of the flow field using tailored finite difference codes specific to the experiments. We will investigate two important cases, stretching and folding of the interface for dilute systems with finite jump in viscosity, and the bifurcation from diffusive to convective mass transport. Apart from its microgravity relevance, this experiment is well suited to study dynamics in nonlinear systems.		

Task Description:	The objective of our research is to study mixing of two fluids generated by steady acceleration in order to understanding transport phenomena in a microgravity environment with application to g-jitter acceleration. We propose to carry out ground-based and eventually flight experiments to quantify flow fields due to g-jitter type of accelerations using Stereo Particle Imaging Velocimetry (PIV), and quantify the concentration field using image processing techniques. The understanding of the effects of steady-g and g-jitter on transport phenomena is of great practical interest to the microgravity community and impacts the design of experiments for the Space Shuttle as well as the International Space Station. The aim of our proposed research is to provide quantitative data to the community on the effects of steady and g-jitter accelerations on flow fields due to mixing induced by buoyancy forces. This research work will impact our understanding of the effects of buoyancy induced flow fields on fluid mixing, generated by steady and g-jitter accelerations. This is important because the fundamental phenomenon of mixing occurs in a broad range of materials processing encompassing the growth of opto-electronic materials and semiconductors, (by directional freezing and physical vapor transport), to solution and protein crystal growth. The effect of the governing parameter space (Gr, Sc, Ar, and Re) on the kinematics of the interface will be thoroughly studied, which will allow differentiation between convection and diffusion dominated regions. The basic structures of mixing, whorls and tendrils, will be identified for our system. The global topology of the flow field will yield insight in the fundamental cause of mixing, i.e., homoclinic and heteroclinic points, and horseshoe maps. In addition, the experiments will provide PIV flow field data for comparison with computational models.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this reporting period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)